Lambda Functions in Python

Lambda Function :

- A **lambda function** is a **small anonymous function** in Python.
- It is used to create small, **throwaway functions** without formally defining them using `def`.

Anonymous Function means a function **without a name**.

Syntax of Lambda Function

```
lambda arguments: expression
```

- Can have **any number of arguments**, but **only one expression**.
- The result of the expression is **automatically returned**.

Example 1: Simple Lambda Function

```
square = lambda x: x * x
print(square(5))   # Output: 25
```

Explanation:

- `lambda x: x * x` creates an anonymous function that returns square of x.
- It is assigned to variable `square`.

Example 2: Lambda with Multiple Arguments

```
python

add = lambda a, b: a + b
print(add(10, 20))   # Output: 30
```

Lambda vs Regular Function

Regular Function	Lambda Function
Uses `def` to define function	Uses `lambda` keyword
Can have multiple expressions	Only **one** expression
Can have a name	Usually **anonymous**
More readable for complex logic	Suitable for **small, short functions**

Use Cases of Lambda Functions

Lambda functions are commonly used with:

1. `filter()` – filter items based on a condition
2. `map()` – apply a function to all elements
3. `reduce()` – reduce elements to a single value